

ARMELLE DUSTON

Boston, Massachusetts

☎ 434-851-0200 ✉ aduston@g.harvard.edu 🌐 [linkedin.com/in/armelle-duston-88b498197](https://www.linkedin.com/in/armelle-duston-88b498197)

EDUCATION

Harvard T.H.Chan School of Public Health

August 2024 – Present

PhD Biostatistics

Vrije Universiteit Amsterdam

August 2022 – December 2022

Semester Abroad

Colorado School of Mines

August 2020 – May 2024

BS Applied Mathematics and Statistics

GPA: 4.0

RELEVANT COURSEWORK

- Probability I
- Inference I/II
- Statistical Methods I/II
- Computing I
- Spatial Statistics
- Time Series Analysis
- Multivariate & Longitudinal Analysis
- Bayesian Methods
- Causal Inference
- Intro to Epidemiology
- Environmental Health

EXPERIENCE

Harvard T.H. Chan School of Public Health

January 2025 – Present

Doctoral Research with Drs. Brent Coull and Rachel Nethery

Boston, MA

- Studied diverse spatial methods with a focus on Bayesian disease mapping models.
- Worked on statistical model development to account for population count errors in US Census data in the context of small-area spatial analysis.

Colorado School of Mines

May 2022 – May 2024

Undergraduate Research with Dr. Cecilia Diniz Behn

Golden, CO

- Studied circadian health in adolescents using statistical and computational modeling techniques.
- Worked with experimental and quantitative/mathematical collaborators.
- Presented research to sleep-science and mathematical audiences.
- Designed and presented posters and presentations at several conferences and first-authored a manuscript titled "Sex differences in circadian timing and biological night in adolescents."

Colorado School of Mines

January 2022 – May 2023

Undergraduate Research with Dr. Michael Mikucki

Golden, CO

- Wrote Python software to perform automated sensitivity analysis for dynamical systems (ordinary differential equations)
- Used the software to investigate SIR (infectious disease) models and other dynamical systems.
- Designed and presented posters at several conferences.

Summer Institute in Biostatistics at NC State & Duke

June 2023 – July 2023

Participant

Raleigh, NC

- Analyzed medical data using statistical techniques in SAS and R, met practicing physicians and biostatisticians engaged in medical research, and learned about principles of applied biostatistics including Bayesian statistics, causal inference, and survival analysis.
- Competed in the data hackathon culminating the program in which we cleaned and organized a database of 110 predictors for myocardial infarction relapse and built and evaluated machine learning models in R.

Applied Math & Stats Tutoring Center

August 2023 – December 2023

Tutor

Golden, CO

- Helped students with advanced math topics such as linear algebra and differential equations by breaking down math concepts and helping students use tools like visual aids to support their learning.

TECHNICAL SKILLS

Programming Languages/Technology: R, Python, MATLAB, LaTeX, SQL, C++, GitHub

PUBLICATIONS

- **Duston, A.**, Holtman, S., Bowen, A. E., Cree, M. G., Nadeau, K., Wright Jr., K. P., Simon, S. L., & Diniz Behn, C. G. (2025). Sex Differences in Circadian Timing and Biological Night in Adolescents. *Journal of Biological Rhythms*, 40(1), 7–18. <https://doi.org/10.1177/07487304241309165>
- Stowe, S. R., **Duston, A.**, Robinson, W., & Diniz Behn, C. (2025). Analyzing the Interactions of Light and Melatonin Forcing in a Mathematical Model of the Human Circadian Oscillator. *Journal of Pineal Research*, 77(3), e70056. <https://doi.org/10.1111/jpi.70056>

PRESENTATIONS

Nebraska Conference for Undergraduate Women in Mathematics (Poster) **February 2024**

- Don't PASA your time writing down sensitivity equations! Automated sensitivity analysis for dynamical systems

Mines Undergraduate Research Symposium (Poster) **May 2023**

- Don't PASA your time writing down sensitivity equations! Automated sensitivity analysis for dynamical systems

CU Anchutz Women's Health Research Day (Poster) **April 2023**

- Sex differences in the circadian timing of adolescents

Nebraska Conference for Undergraduate Women in Mathematics (Slides) **February 2023**

- The relationship between adolescent obesity and sleep

Colorado Sleep and Circadian Summer School (Slides) **August 2022**

- The relationship between adolescent obesity and sleep

HONORS AND AWARDS

Outstanding Graduate in Applied Math & Statistics **March 2024**

Top student award given by AMS department

Colorado School of Mines

Undergraduate Research Distinction **May 2024**

Recognizes significant research achievements

Colorado School of Mines

Harvey Scholarship **August 2020 – May 2024**

Merit-based full tuition scholarship

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Undergraduate Research Fellowship **August 2022 - May 2023**

Funds undergrad research projects

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CLASS PROJECTS

Spatial Statistics: Modeling the Relationship Between Air Quality and Lung Cancer **December 2023**

- Used EPA and National Cancer Institute datasets and spatial statistical tools in R to evaluate air quality as a predictor for rates of lung cancer in the US and investigated relationship between those factors and sociodemographic status. Visualized the data, predictions, and errors associated with those predictions on a US map.

Comp. Neuroscience: Relating MAO and TH Dynamics in Schizophrenia to Dopamine and Sleep **May 2023**

- Investigated the relationship between the circadian clock and schizophrenia using a model of differential equations describing the molecular clock's effects on the dopaminergic system and altered the dynamics of enzymes Monoamine Oxidase (MAO) and Tyrosine Hydroxylase (TH) according to known behaviors in people with schizophrenia.

EXTRACURRICULAR

Mathematical Biology Research Group **May 2022 – May 2024**

Member

Colorado School of Mines

Society of Women in Mathematics **May 2022 – May 2024**

Member

Colorado School of Mines